A Naive Linear Regression Perspective on the Cases in Faulhaber's Formulae

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Abstract

The well-known Faulhaber's Formulae, discovered by J. Faulhaber (1631) and found in Knuth (1993), give a handy generalisation of the sum of conseuctive natural integers all raised to an arbitrary power of certain natural integers; in fact, the special cases of power 1 and 2 are so well-known that they are printed across textbooks in some mathematical training institutions. However, in this article, we try to use a naive method of solving simple (yet sometimes tedious) systems of linear equations to acquire formulae equivalent to the aforementioned.

1 Stimulus

Let us revisit the Faulhaber's Formulae [2], so as to ensure that we are on the same page. We define that

$$\forall m \in \mathbb{N}, \quad \sum_{j=0}^{m} \binom{m+1}{j} B_j^+ = m+1,$$

and that

$$B_n^- = (-1)^n B_n^+$$

to make it alternating in sign, and call that B_n^- the n-th **Bernoulli number**. As a result, the defined Bernoulli numbers come in the sequence of

$$1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \ldots.$$

In addition, even in B_j^-, the convention states that instead of $-\frac{1}{2}$, the first Bernoulli number should be $\frac{1}{2}$. The Bernoulli numbers are used in the generic, closed form of the Faulhaber's Formulae, written as

Theorem 1 (Faulhaber's Formulae)**.**

$$\sum_{k=1}^{n} k^m = \frac{1}{m+1} \sum_{j=0}^{m} \binom{m+1}{j} B_j^- n^{m-j+1}$$

Proof. Omitted since this is not the focus of the article. In fact, a detailed and rigorous proof can be found at any texts dedicated to the development and consequences of the Formulae, examples include [1]. □

In particular, here are some *really common* and 'handy' (*cf.* the Abstract) consequences of the Formulae of smaller exponents:

$$\begin{aligned}
\sum_{k=1}^{n} k &= \frac{\left(n^2+n\right)}{2} \\
\sum_{k=1}^{n} k^2 &= \frac{\left(2n^3+3n^2+n\right)}{6} \\
\sum_{k=1}^{n} k^3 &= \frac{\left(n^4+2n^3+n^2\right)}{4} = \left(\frac{\left(n^2+n\right)}{2}\right)^2 \\
\sum_{k=1}^{n} k^4 &= \frac{\left(6n^4+15n^4+10n^3-n\right)}{30} \\
\sum_{k=1}^{n} k^5 &= \frac{\left(2n^6+6n^5+5n^4-n^2\right)}{12}
\end{aligned}$$

2 Lemmas and Development

However, this sounds a little bit too **analytical** (as in, mathematical analysis). We would like to reduce the *proofs* of the formulae into elementary forms, so as to reach a broader audience demanding a milder list of mathematical prerequisites. In simpler words, we would like to rely on some methods for solving in linear algebra without the need of deriving such formulae. To do this, let make an observation to the equations above, first produced by the Faulhaber's formulae and not ours (yet). We surmise:

Theorem 2. *The sum of consecutive natural numbers, from* 0 *to* x*, with each to the degree* $n \in \mathbb{N}$*:*

$$\sum_{q=0}^{x} q^n = P_n[x]$$

gives polynomials in x *of strictly* $n+1$ *degrees.*

Rest assured, it is always true. However, even proving this mere property of existence is by no means trivial. We will do an induction, since this method is cleaner and more intuitive.

Proof. Without loss of generality let us shift the index from 0 to 1. Indeed, $\forall k \in \mathbb{R} \neq 0, 0^k = 0$ and $\forall k \in \mathbb{R}, 0 + k = k$. Without digressing further to abstract algebra for now, let us focus on the main proof.

Before the induction, we actually need a special expression that will be handy in the rest of the proof. We begin with the aid of the binomial theorem, which dictates that the expansion of $(q+1)^{n+1}$ (treat q as any integer and ignore the sigma in the statement of the theorem temporarily, first) must be

$$(q+1)^{n+1} = \binom{n+1}{0} q^{n+1} + \binom{n+1}{1} q^n + \binom{n+1}{2} q^{n-1} + \cdots + \binom{n+1}{n} q + \binom{n+1}{n}.$$

Then move the term q^{n+1} from right to left yields

$$(q+1)^{n+1} - q^{n+1} = \sum_{w=0}^{n} \binom{n+1}{w} q^w.$$

If we sum both sides of this identity from $q = 1$ to x, then it gives the formula

$$\sum_{q=1}^{x} \left((q+1)^{n+1} - q^{n+1}\right) = \sum_{q=1}^{x} \sum_{w=0}^{n} \binom{n+1}{w} q^w$$

We would like to try out the left-hand-side first, since it looks less horrible. Surprisingly, substituting, we discover that

$$\sum_{q=1}^{x} \left((q+1)^{n+1} - q^{n+1}\right) = (2^{n+1} - 1^{n+1}) + (3^{n+1} - 2^{n+1}) + \cdots + ((x+1)^{n+1} - x^{n+1})$$

It telescopes to an elegant and simple $(x+1)^{n+1} - 1$. It then gives

$$(x+1)^{n+1} - 1 = \sum_{q=1}^{x} \sum_{w=0}^{n} \binom{n+1}{w} q^w.$$

We, however, still need a special term to cater for our inductive hypothesis, namely the term q^n and its coefficient. Isolating this very term allows us to view the identity in a different lens. At first glance in the intimidating right-hand-side this term seems to be nowhere to be found. However, upon closer scrutinisation, we find out that the term

$$\binom{n+1}{1} q^n$$

is secretely hiding in the generating function from our binomial expansion. Thus, after we have done some index-shifting work, we arrive at

$$\sum_{q=1}^{x} \binom{n+1}{1} q^n + \sum_{q=1}^{x} \sum_{w=0}^{n-1} \binom{n+1}{w} q^w.$$

Expanding $\binom{n+1}{1} = \frac{(n+1)!}{1!n!}$ gives $(n+1)$, and hence we arrive at

$$(n+1) \sum_{q=1}^{x} q^n + \sum_{q=1}^{x} \sum_{w=0}^{n-1} \binom{n+1}{w} q^w.$$

To our surprise, thanks to the discrete Fubini's principle, we can exchange the double sigmas to retrieve

$$(n+1) \sum_{q=1}^{x} q^n + \sum_{w=0}^{n-1} \left(\binom{n+1}{w} \sum_{q=1}^{x} q^w \right).$$

Define a shorthand for the expression

$$\sigma_\kappa = \sum_{q=1}^{x} q^\kappa, \quad \forall \kappa \in \mathbb{N}.$$

Then, we are able to substitute this shorthand into the above expression and turn them into a nicer- and more compact-looking identity:

$$(x+1)^{n+1} - 1 = (n+1)\sigma_n(x) + \sum_{w=0}^{n-1} \binom{n+1}{w} \sigma_w(x).$$

We then finally isolate the σ_n and turn them into a recurrence relation using some elementary algebra:

$$\frac{1}{n+1}\left((x+1)^{n+1} - 1 - \sum_{w=0}^{n-1} \binom{n+1}{w} \sigma_w(x)\right) = \sigma_n(x).$$

With this convenient definition on hand, we may finally begin our **strong induction**.

The basic case can be proved way more concisely and independently from the inductive case when $n = 0$, by letting

$$1^0 + 2^0 + \cdots + x^0 = \sum_{n=1}^{x} 1 = x.$$

It itself is already a linear monomial. However, for the sake of consistency, let us utilise the recurrence relation we just worked out:

$$\sigma_0(x) = \frac{1}{0+1}\left((x+1)^1 - 1\right) = x,$$

where results concur.

The inductive case is somewhat more challenging. We first assume that for all integers w when $0 \leq w < n$, the sum $\sigma_j(x)$ is a polynomial in x of degree $w+1$. Mathematically,

$$\forall w \in \mathbb{Z} : w \in [0, n), \quad \sigma_w(x) \in \mathbb{R}_{w+1}[x].$$

Analysed based on the derived expression, the term $(x+1)^{n+1} - 1$ is already in $\mathbb{R}_{n+1}[x]$ (we will use the polynomial ring notation for the sake of clarity). Furthermore, by the inductive hypothesis, $\forall \sigma_w(x) \in \mathbb{R}_{w+1}[x]$. We also notice that only elementary arithmetics such as subtraction of the $\sigma_w(x)$ terms, division by $n+1$ in the expression. All these operations do not change the nature of being a polynomial. Even better, since the highest degree amongst the inductive terms are $((n-1)+1) = n$, the presence of solely basic arithmetics ensures that the x^{n+1} term remains intact, preserving the nature of its degree (so is the coefficient of the reciprocal of $(n+1)$).

From the principle of induction, $\sigma_n(x)$ must be in $\mathbb{R}_{n+1}[x]$. (The operations we have initially been investigating pertain strictly to positive integers only, leaving the imaginary part $\mathfrak{Im}(\sigma_n(x)) = 0$.) □

Since we are convinced that the sums always give a polynomial regardless (but depending on n), we already have a clear idea of a multiple linear regression (i.e., linear regression but with multiple variables and with a system of equations). We set systems of linear equations in the form:

$$a_{n,1}(k)^{n+1} + a_{n,2}(k)^n + \cdots + a_{n,n+1} \equiv \sum_{i=0}^{n} a_{n,(n+2)-i}(k)^i = s_{k,n} \tag{1}$$

Where a denotes a variable (ironically, k, which is to denote the variable x in the polynomial function we eventually will work out to output the sum till x, is a constant at this moment) k denotes the k-th positive integer, and n the corresponding exponent (consistent with the previous notation). For example, $1^2+2^2+3^2 = 14$ can be expressed as

$$a_{2,1}(3)^3 + a_{2,2}(3)^2 + a_{2,3}(3) + a_{2,4} = 14.$$

We choose to start from 1 all the way until we created a suitable number of linear equations to ensure a solution. How many equations in this form do we require?

Theorem 3. *Informally, for systems of linear equation consisting of n variables having a unique solution, there exist at least n different formulae to represent it, with no two equations are exactly the same or differ only by the constant term.*

Proof. We utilise the Cramer's Rule, which state that the system of n linear equations of n variables expressed by the $n \times n$ coefficient matrix A, the $n \times 1$ column vector of variables $\vec{x}$ and the $n \times 1$ column vector of constants $\vec{b}$, in the form of

$$A\vec{x} = \vec{b},$$

and that $\det(A) \neq 0$, gives any specific variable $x_i, 1 \leq i \leq n$ in the form of

$$x_i = \frac{\det(A_i)}{\det(A)} \tag{2}$$

Where A_i is the matrix formed by replacing the i-th column of A with the constant column $\vec{b}$. Since there are n variables, we can generate a unique structural formula for each by each:

- $x_1 = \dfrac{\det(A_1)}{\det(A)}$
- $x_2 = \dfrac{\det(A_2)}{\det(A)}$
- $\cdots$
- In general, for x_n, have the i in the Equation (2) replaced.

Each formula x_i is functionally and structurally different because the matrix A_i in the numerator is altered at i for each variable. Since it is solvable, modifying different columns alters the resulting determinant calculation uniquely for each variable, resulting in the demand for n unique formulae to solve. □

This means, in this case, to obtain a solution from $a_{n,1}$ to $a_{n,(n+2)-i}$, we need $(n+2)-0 = (n+2)$ formulae (i has a minimum as defined at 0). However, it turns out to be unnecessary:

Theorem 4. *x is a factor of any polynomial $\sigma_n(x)$.*

Proof. By convention,

$$\sum_{q=0}^{0} q^n = 0$$

Even without the rational root theorem have we already been able to mentally-substitute. □

Then we can eliminate the final $a_{n,n+2}$, or rather, it is always 0. This zero is drawn from the fact that the last variable does not attach to any substitution (but instead, the substitution raised to 0).

3 The Actual Method

Since we established the fact that $\sigma_n(x) \in \mathbb{R}_{n+1}[x]$ and derived Equation (1), it is high time we utilised them accordingly. To be precise, we will pre-compute the sums from 1 to $n+1$, and substitute them back into the equation. The author has accumulated so much ado till now, and candidly, our goal is but *to find the coefficients* of the polynomials in the above non-constructive proof through setting up equations, which require the verified values to begin with (in other words, another additional method for this preliminary computation is required, such as brute-forcing or, ironically, the Faulhaber's Formulae). Indeed, since we know that the variables and the return values are true, we may as well have the slots of the variables occupied, then solve using linear algebra methods.

Remark 1. It is sad where an article titled 'Regression' can hardly mention statistics. Here it comes, but just in a few hasty words. This 'misleadingness' is because of the fact that this article mostly centres on the leisurely utilisation of linear algebraic methods, and statistics is merely as a tool to confirm the validity of the retrieved value only (in this article only. Apologies). Since we proved that there is a solution and we supply the system with just enough formulae describing the desired tuple of answers, then it is not hard to perceive why the $R^2 = 1$ since every single point (on regression, in the form of $k = s_{k,n}$) is passed through.

Using the Cramer's Rule or Gaussian elimination (or whichever valid method) for larger matrices or the determinant-adjugate method for smaller matrices themselves again, we can draw solutions from $a_{n,1}$ to $a_{n,n+1}$ by doing the tremendously tedious Laplacan determinant calculations, dividing and conquering and so on and so forth.

Here we bump into a small demonstration serving as an alternative way (besides the hook – the Faulhaber Formulae) to retrieve the sum when $n = 1$:

Example 1. We first compute that $1^1 = 1$ and $1^1 + 2^1 = 3$. With this result, alongside the powering processes given in Equation (1), we arrive at (and would like to solve)

$$\begin{cases} a_{1,1} + a_{1,2} = 1 \\ 4a_{1,1} + 2a_{1,2} = 3 \end{cases}$$

With a bit of linear algebra, specifically setting up the matrix and vectors

$$\begin{bmatrix} 1 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} a_{1,1} \\ a_{1,2} \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

These matrices are still reasonable to handle using the determinant-adjugate form. Consider the fact that

$$\begin{bmatrix} a_{1,1} \\ a_{1,2} \end{bmatrix} = \frac{1}{(1)(2)-(1)(4)} \begin{bmatrix} 2 & -1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

doing scalar and matrix multiplications yields

$$A^{-1} = \begin{bmatrix} -1 & \frac{1}{2} \\ 2 & -\frac{1}{2} \end{bmatrix}.$$

Solving should retrieve that

$$\begin{bmatrix} a_{1,1} \\ a_{1,2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix},$$

yielding

$$\sigma_1(x) = \frac{1}{2}x^2 + \frac{1}{2}x.$$

This matches the well-known

$$\sigma_1(x) = \frac{(x+1)x}{2}.$$

Example 2. Let us do a larger one with bolder audacity, say, when $n = 5$. Compute:

- $1^5 = 1$.
- $1^5 + 2^5 = 33$.
- $1^5 + 2^5 + 3^5 = 276$.
- $1^5 + 2^5 + 3^5 + 4^5 = 1\,300$.
- $1^5 + 2^5 + 3^5 + 4^5 + 5^5 = 4\,425$.
- $1^5 + 2^5 + 3^5 + 4^5 + 5^5 + 6^5 = 12\,201$.

Then, we set up the lines accordingly (with reference to Equation (1)):

$$\begin{cases} a_{5,1} + a_{5,2} + a_{5,3} + a_{5,4} + a_{5,5} + a_{5,6} = 1 \\ 64a_{5,1} + 32a_{5,2} + 16a_{5,3} + 8a_{5,4} + 4a_{5,5} + 2a_{5,6} = 33 \\ 729a_{5,1} + 243a_{5,2} + 81a_{5,3} + 27a_{5,4} + 9a_{5,5} + 3a_{5,6} = 276 \\ 4\,096a_{5,1} + 1\,024a_{5,2} + 256a_{5,3} + 64a_{5,4} + 16a_{5,5} + 4a_{5,6} = 1\,300 \\ 15\,625a_{5,1} + 3\,125a_{5,2} + 625a_{5,3} + 125a_{5,4} + 25a_{5,5} + 5a_{5,6} = 4\,425 \\ 46\,656a_{5,1} + 7\,776a_{5,2} + 1\,296_{5,3} + 216a_{5,4} + 36a_{5,5} + 6a_{5,6} = 12\,201 \end{cases}$$

It looks daunting, but luckily we got Wolfram Alpha (disclaimer: not sponsored). After running through it in the computer, we get

$$\vec{x} = \begin{bmatrix} \frac{1}{6} \\ \frac{1}{2} \\ \frac{5}{12} \\ 0 \\ -\frac{1}{12} \\ 0 \end{bmatrix}.$$

Extract $\frac{1}{12}$ and simplify, we get the familiar $\frac{(2n^6 + 5n^5 + 5n^4 - n^2)}{12}$ derived earlier using the Faulhaber alternative.

4 Remarks and Caveats

(First, apologise for some Algebra seeping into this section.) As mentioned, we only went through the cases when $n \in \mathbb{N}$. Therefore, it would largely be only valid in such range. (Agreeing with our common sense: from the cardinalities' perspective, if n were allowed to $\in \mathbb{Q}$, it would be a fifty-fifty for one formula producing a sensible result: a closed integer, thanks to $\mathbb{Z}$; and an absurd rational (we are summing up whole numbers so it is to be closed!), thanks to $\mathbb{Q}$. Even worse, if n were allowed to $\in \mathbb{R}$, since $|\mathbb{R}| > |\mathbb{Q}| = |\mathbb{Z}|$, it would be ridiculous to just think about that it would be almost impossible to get a nice, albeit incorrect rational out of the overwhelming transcendentals or irrationals. **Domain is important.**)

Another caveat revolving around the importance of the domain stands as the fact that the generators of the coefficients – or without Algebra jargon, the number we chose as the base for the exponentiation –

Theorem 5. *For the aforementioned system of linear equation to be valid, $n \in \mathbb{Z} > 0$.*

Proof. **When $\mathbf{n < 0}$**, not only the exponentiation produces smaller and smaller rationals (fractions, for the sake of consistency), but more importantly, it contradicts with our definition of the domain, where x must range from 0 onwards. Producing the constant term at the right is difficult, since there is arrantly no definition whatsoever. Another way of viewing this is the use of $\frac{1}{n+1}$ when we did our induction in the beginning. Obviously this expression is undefined when $n = -1$; however, successfully going through the induction (to even '*topple the Domino*' successfully), the closed definition of such expression is explicitly required.

When $\mathbf{n = 0}$, $\sum_{q=0}^{x} 0q = 0$ and consequently

$$0 = 0$$

is, unfortunately and vulgarly, useless, in solving the system of equation. Then, it is destined to be not solvable, due to the theorem regarding the number of unique equations. □

Remark 2. Although this algorithm can be practical and useful in certain areas, it is not nearly perfect (and thus, *Naive*). Its main limitation is being somewhat slow, since the preparation works are tremendous. The prerequisites of this substitution involve finding the sum of each polynomial first (boiling down to $\mathcal{O}(n)$), then find the corresponding exponents (to $\mathcal{O}(n)$) and the arduous Gaussian elimination (to $\mathcal{O}(n^3)$). Absorbing, $\mathcal{O}(n^3)$ stands as the time complexity of this method. This is way slower than even the 'primitive'-sounding brute-forcing, which gives an asymptotic time complexity of logarithmic $\mathcal{O}(\log n)$ if the fast exponentiation method is adopted, or still linear $\mathcal{O}(n)$ even though n is treated as a constant. Worse still, to execute the algorithm described in the hook, the Faulhaber Formulae with unknown Bernoulli numbers, takes $\mathcal{O}(n^2)$ time only.

Remark 3. Instead, the method explored in this article serves as a demonstration of the unexpected accessible possibility of crossing linear algebra over elementary number theory. Perhaps the rigour of the proofs, the perception of the beauty of mathematics and the exploration itself would be more important and enjoyable than heavy, practical real-world usage.

Remark 4. Of course, perhaps for programmes or occasions where other alternatives may be too tedious to adopt and devise, this method may serve as an effective alternative on machines, say, with only `scikit-learn` available.

Remark 5. We may as well draw a nice conclusion here from our excursion that is supposed to be leisurely and understandable, despite the author's personal heavy use of LaTeX, as well as the complex-looking formulae.

References

[1] Donald E. Knuth. 'John Faulhaber and sums of powers'. In: *Mathematics of Computation* 61.203 (1993), pp. 277–294.

[2] Eric W. Weisstein. *Faulhaber's Formula*. MathWorld – `https://mathworld.wolfram.com/FaulhabersFormula.html`. Accessed: 7 August 2026. n.d.